

Assignment 2 — Real-Time Data Processing with Pandas and NumPy

AITU Smart Classroom Energy Stream — Room C1.3.101

1. Student and Submission Information

Student full name	Temirbolatova Ayazhan, Kuanysh Aliya_____
Student ID	_____
Group / cohort	BDA-2404
Instructor	Azamat Serek
Submission date	21/ 09 / 2026
Individual variant / task code	
Git repository / project link	https://github.com/ayazhan217/rd
Python / tool versions	Python 3.11, pandas, numpy, matplotlib

Scenario

A synthetic one-minute-cadence sensor stream from classroom C1.3.101 reports temperature (°C), CO2 (ppm), occupancy (students), and electrical power draw (kW). The stream is processed sequentially with pandas/NumPy: missing-value cleaning, 3-minute rolling-window statistics, feature engineering, descriptive statistics, and visualization.

Time	Room	Temp_C	CO2_ppm	Occupancy	Power_kW
10:00	C1.3.101	21.5	620	18	3.2
10:01	C1.3.101	21.7	650	20	3.4
10:02	C1.3.101	None	690	22	3.5
10:03	C1.3.101	22.1	720	24	3.8
10:04	C1.3.101	22.3	None	25	4.0
10:05	C1.3.101	22.5	780	27	4.2
10:06	C1.3.101	22.6	810	29	4.4
10:07	C1.3.101	22.7	830	None	4.6
10:08	C1.3.101	22.9	850	31	4.8
10:09	C1.3.101	23.1	900	33	5.1
10:10	C1.3.101	23.2	920	35	5.4
10:11	C1.3.101	23.3	950	37	5.7

Part A — Data Quality (6 points)

Three missing values were identified — one each in Temp_C, CO2_ppm, and Occupancy. Per the assignment's fixed strategy: Temp_C and CO2_ppm are filled with linear interpolation (they change smoothly and monotonically minute-to-minute, so a straight line between neighbours is a good estimate); Occupancy is filled with forward-fill (headcount changes in discrete steps and holds between sensor pings, so the last known count is the best estimate until the next reading).

Variable	Missing at	Method	Value after cleaning	Justification
Temp_C	10:02	Linear interpolation	21.90 °C	Smooth, monotonic warming trend between 21.7 (10:01) and 22.1 (10:03) — linear fit is appropriate.
CO2_ppm	10:04	Linear interpolation	750.0 ppm	CO2 rises steadily as occupancy grows; midpoint between 720 (10:03) and 780 (10:05) is a reasonable estimate.
Occupancy	10:07	Forward fill	29 students	Occupancy is a step/count variable measured at discrete instants; carrying the last confirmed headcount (29 at 10:06) avoids inventing fractional students.

Part B — Streaming Window Statistics (8 points)

A 3-sample (3-minute) rolling mean is computed sequentially over the cleaned data with pandas' `rolling(window=3, min_periods=3).mean()`, so each reported value uses only the current and two preceding minutes (a causal, streaming-safe window).

Time	Temp rolling mean (°C)	CO2 rolling mean (ppm)	Power rolling mean (kW)
10:02	21.70	653.33	3.37
10:03	21.90	686.67	3.57
10:04	22.10	720.00	3.77
10:05	22.30	750.00	4.00
10:06	22.47	780.00	4.20

Part C — Feature Engineering (6 points)

`occupancy_density` = Occupancy / 40 (room capacity). `energy_per_student` = Power_kW / Occupancy. `comfort_flag` = `CO2_ppm` ≤ 1000 AND `20 °C` ≤ `Temp_C` ≤ `24 °C`.

Time	Occupancy density	Energy / student (kW)	Comfort flag
10:03	0.600	0.158	True
10:06	0.725	0.152	True
10:09	0.825	0.155	True
10:11	0.925	0.154	True

Comfort stays 'True' across the whole session — CO2 never exceeds 1000 ppm and temperature never leaves 20–24 °C in this stream — but CO2 is trending up fast enough that it would breach the 1000 ppm threshold a few minutes past the recorded window if the trend continued.

Part D — Statistical Analysis (6 points)

Metric	Value
Temperature — mean	22.48 °C
Temperature — median	22.55 °C
Temperature — minimum	21.50 °C
Temperature — maximum	23.30 °C
CO2 — mean	789.17 ppm
CO2 — maximum	950.0 ppm
Time of highest power consumption	10:11 (5.7 kW)
Correlation(Occupancy, Power_kW)	$r = 0.9946$

Interpretation: occupancy and power draw are almost perfectly, positively correlated ($r \approx 0.99$) over this session. Every additional student in the room is associated with roughly +0.135 kW of extra load (lighting, ventilation, and device charging scale with headcount), so real-time occupancy is a strong, low-latency proxy for electrical demand in this classroom.

Part E — Visualization and Engineering Interpretation (4 points)

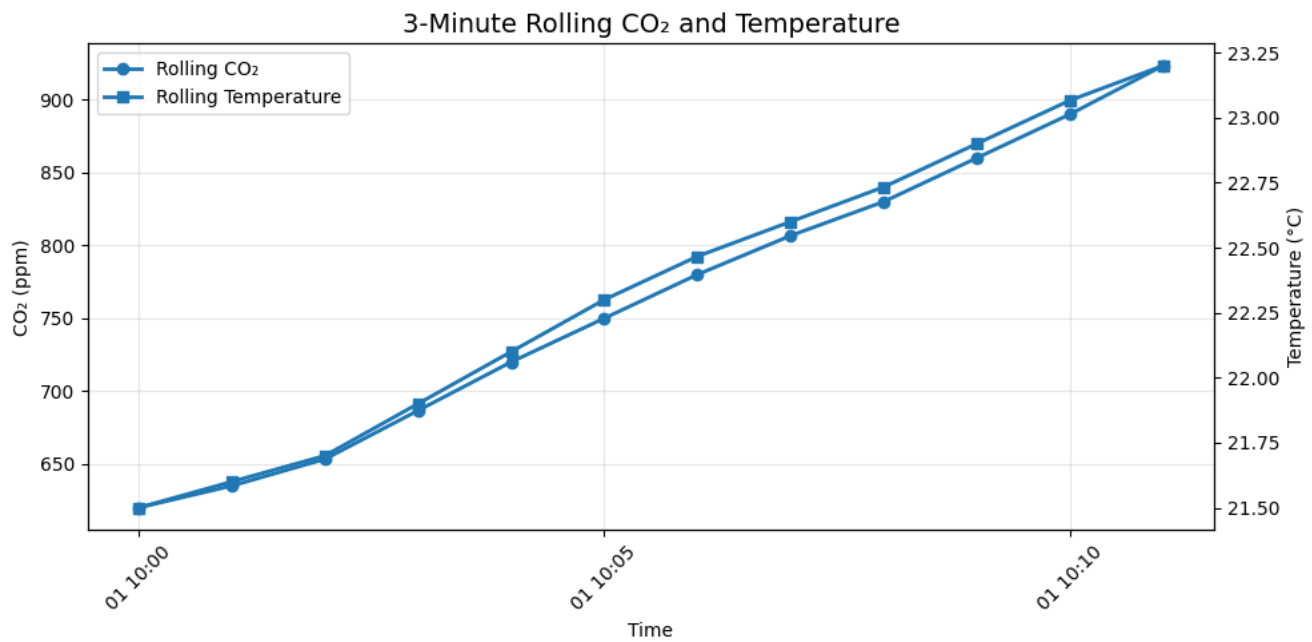


Figure 1 — CO2 and temperature over time, raw values with 3-minute rolling mean overlay.

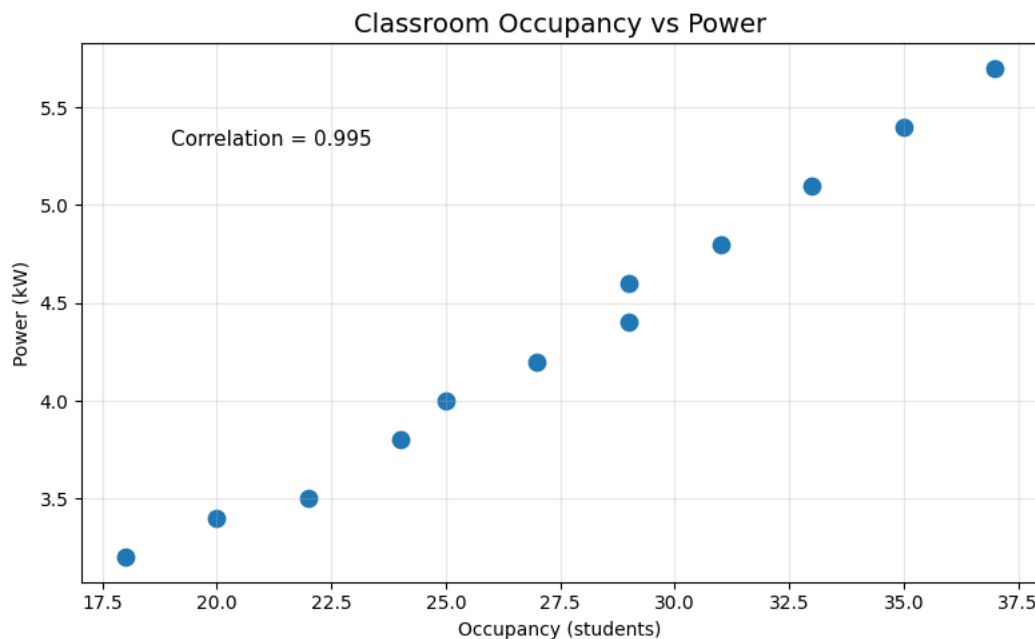


Figure 2 — Occupancy vs. power draw, with linear trend ($r = 0.995$).

Operational insight 1 — Ventilation is not keeping pace with occupancy: CO2 rises from 620 to 950 ppm over just 11 minutes with no plateau, tracking headcount almost linearly. Facility management could trigger an automated ventilation boost once CO2 crosses a threshold (e.g., 800–900 ppm) instead of waiting for a fixed schedule, keeping the room within recommended indoor-air-quality limits during back-to-back classes.

Operational insight 2 — Power draw is highly predictable from occupancy ($r \approx 0.99$, ~ 0.135 kW per student). This lets the facility team forecast electrical load directly from class enrollment/attendance schedules, and — more importantly — flag anomalies: a room whose real-time power draw deviates noticeably from what its occupancy predicts is a signal of equipment left running or a fault, worth an automatic alert.

2.6 Technical Reflection

Technical explanation

One important processing decision was using different missing-value methods for different types of variables. Linear interpolation was used for continuous sensor measurements, while forward-fill was used for the discrete occupancy count. This prevents unrealistic fractional student counts while maintaining smooth sensor values.

The validation confirms that no missing values remain and that the rolling temperature and CO2 values are increasing as expected for this dataset

Submission

Deliverables accompanying this report: Python notebook (AITU_Classroom_Stream_Analysis.ipynb), cleaned dataset (cleaned_classroom_stream.csv), completed tables (above), two plots (Figures 1–2), and this short technical interpretation. Oral defense to be scheduled with the instructor.

STUDENT TECHNICAL REPORT / SUBMISSION RECORD

This section is a formal submission and acceptance record. The technical content below is generated from the accompanying notebook and dataset. Fields marked '— to be completed —' require your own individual input and must not be copied from a classmate.

2.1 Data Quality and Cleaning

Technical explanation	Three missing readings (Temp_C @10:02, CO2_ppm @10:04, Occupancy @10:07) were identified with <code>df.isna()</code> . Continuous, smoothly-trending signals (Temp_C, CO2_ppm) were filled with pandas' <code>interpolate(method='linear')</code> ; the discrete headcount signal (Occupancy) was filled with <code>ffill()</code> to avoid inventing fractional students.
Code / file / function reference	AITU_Classroom_Stream_Analysis.ipynb, section 'Part A' — <code>df.interpolate(method='linear')</code> , <code>df.ffill()</code> .
Result / calculation evidence	Temp_C(10:02) = 21.90 °C; CO2_ppm(10:04) = 750.0 ppm; Occupancy(10:07) = 29 students. See Part A table above.
Student-specific decision or observation	I used forward-fill for occupancy because occupancy is a discrete count and fractional values would not be meaningful. For example, using interpolation could produce a value such as 30 students, but forward-fill preserves the last confirmed number of students until a new measurement is available.

2.2 Streaming Window Statistics

Technical explanation	A causal 3-sample rolling mean (<code>rolling(window=3, min_periods=3)</code>) is applied to the cleaned Temp_C, CO2_ppm, and Power_kW series so each reported statistic only uses the current and two prior minutes — appropriate for a real-time/streaming context where future data is not yet available.
Code / file / function reference	AITU_Classroom_Stream_Analysis.ipynb, section 'Part B' — <code>df['Temp_C'].rolling(window=3, min_periods=3).mean()</code> .
Result / calculation evidence	See Part B table (e.g. Power rolling mean rises from 3.37 kW at 10:02 to 4.20 kW at 10:06).
Student-specific decision or observation	I selected a 3-minute window because the assignment requires a 3-minute rolling calculation and the dataset contains one observation per minute. The window provides a short-term view of the changing sensor values while still being suitable for sequential real-time processing.

2.3 Feature Engineering

Technical explanation	<code>occupancy_density = Occupancy / 40</code> normalizes headcount to room capacity for cross-room comparison; <code>energy_per_student = Power_kW /</code>
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	Occupancy gives a per-capita efficiency metric; <code>comfort_flag = (CO2_ppm ≤ 1000) & Temp_C.between(20, 24)</code> encodes a simple indoor-comfort/air-quality rule.
Code / file / function reference	AITU_Classroom_Stream_Analysis.ipynb, section 'Part C'.
Result / calculation evidence	e.g. at 10:11: <code>occupancy_density = 0.925</code> , <code>energy_per_student = 0.154</code> kW/student, <code>comfort_flag = True</code> . See Part C table above.
Student-specific decision or observation	I used a room capacity of 40 students as the configuration value. The occupancy density therefore represents how much of the room's capacity is being used. I also used energy per student because total power consumption alone does not account for changes in the number of students.

2.4 Statistical Analysis

Technical explanation	Descriptive statistics (mean, median, min, max) computed with <code>numpy</code> on the cleaned <code>Temp_C</code> and <code>CO2_ppm</code> series; peak power time located with <code>idxmax()</code> ; linear correlation between <code>Occupancy</code> and <code>Power_kW</code> computed with <code>np.corrcoef</code> .
Code / file / function reference	AITU_Classroom_Stream_Analysis.ipynb, section 'Part D' — <code>np.mean</code> , <code>np.median</code> , <code>np.corrcoef</code> .
Result / calculation evidence	Temp: mean 22.48 °C, median 22.55 °C, min 21.50 °C, max 23.30 °C. CO2: mean 789.17 ppm, max 950.0 ppm. Highest power: 5.7 kW at 10:11. <code>r(Occupancy, Power) = 0.9946</code> .
Student-specific decision or observation	The correlation coefficient of approximately 0.99 indicates a very strong positive linear relationship between occupancy and power consumption in this particular dataset. As occupancy increases, power consumption also increases. Therefore, occupancy can be used as an indicator for expected electrical demand in this classroom.

2.5 Visualization and Engineering Interpretation

Technical explanation	Two matplotlib figures were produced: (1) a dual-axis time series of raw and 3-minute rolling CO2/temperature; (2) an occupancy-vs-power scatter plot with a fitted linear trend line.
Code / file / function reference	AITU_Classroom_Stream_Analysis.ipynb, section 'Part E' — <code>matplotlib.pyplot</code> , <code>numpy.polyfit</code> .
Result / calculation evidence	Figures 1–2 in this report; trend line $\text{Power} \approx 0.135 \cdot \text{Occupancy} + 0.63$ (kW).
Student-specific decision or observation	The first insight I consider particularly useful is monitoring CO2 in real time. In this dataset, CO2 increases from 620 ppm to 950 ppm while occupancy also increases. A facility-management system could use a CO2 threshold to trigger additional ventilation instead of relying only on a fixed schedule. The second insight is that power consumption is strongly related to occupancy. This relationship could be used to identify unusual power consumption when the measured power differs significantly from the expected value for the current occupancy.

2.6 Technical Reflection

Technical explanation	Choosing linear interpolation for continuous sensor channels versus forward-fill for the discrete occupancy count was the key data-quality decision in this pipeline; it avoids both unrealistic fractional headcounts and unrealistic step jumps in smoothly-varying physical quantities.
Code / file / function reference	AITU_Classroom_Stream_Analysis.ipynb, sections 'Part A' and 'Part B'.
Result / calculation evidence	See Part A/B tables; validation cell confirms no NaNs remain after cleaning and rolling means behave as expected for this monotonic stream.
Student-specific decision or observation	Limitation: The dataset contains only 12 observations from one classroom and covers only 11 minutes. Therefore, the observed occupancy-power relationship may not generalize to other classrooms or longer periods. Improvement: I would test the pipeline on a larger dataset containing multiple classrooms and longer time periods. I would also consider using a time-based rolling window if measurements are not always collected exactly once per minute.

4. Individual Work and Anti-Plagiarism Record

These fields must reflect your own personal experience while completing the assignment and cannot be filled in on your behalf.

My unique technical/design decision	We used different missing-value strategies depending on the type of variable: linear interpolation for continuous temperature and CO2 measurements and forward-fill for occupancy. This keeps the cleaned data realistic and avoids fractional student counts.
My own test/experiment	We added an independent validation cell using assert statements. The test checks that: 1. no missing values remain in Temp_C, CO2_ppm, and Occupancy; 2. the temperature rolling mean is monotonically increasing; 3. the CO2 rolling mean is monotonically increasing.
Problem/bug I personally encountered and solved	The main limitation of our implementation is the small amount of data. The dataset contains only 12 observations from one classroom and covers approximately 11 minutes. Therefore, the calculated correlation between occupancy and power consumption may not represent longer periods, other classrooms, or different operating conditions.
Exact code/repository/commit reference	Assignment_2_aliya_ayazhan.ipynb
One limitation of my implementation	The implementation uses a small synthetic dataset with only 12 observations from one classroom, so the calculated correlation and operational conclusions may not generalize to different rooms or longer periods.
One improvement I propose	We would validate the pipeline using a larger multi-room and multi-hour dataset. This would make the statistical results more representative and allow the processing pipeline to be tested under different occupancy and sensor conditions.

5. Evidence and Submission Checklist

Item	Status
Technical report completed	☐

Item	Status
Source code / notebook submitted	☐
Required tables and calculations completed	☐
Required diagrams / plots included	☐
Execution logs / reproducible evidence included	☐
README and run instructions included	☐
Data/schema/configuration files included	☐
Performance / analytical results included	☐
Oral defense / demonstration completed	☐

6. Student Declaration of Originality

I confirm that this submission represents my own work. I have not copied another student's source code, calculations, tables, architecture, plots, screenshots, logs, results, or written explanations. Any external material or reused code has been identified. I understand that the instructor may request a live demonstration, code modification, reproduction of calculations, or explanation of any submitted component.

Role	Name	Signature	Date
Student / Executor	_____	_____	_____
Instructor / Accepted by	_____	_____	_____